

MTH 401: Equivalence and Absolute Value

Problems 16, 19(b), 20(a), 21

Theresa Wunderlich

1.3 Order and Absolute Value

Although the real numbers satisfy the field axioms given in the previous section, these algebraic properties don't fully characterize $\mathbb{R}$; one thing that's missing is the notion of *order*. We begin by introducing the concept of positive and negative numbers.

Axiom 1.4 (Order). There exists a subset of the real numbers called the set of **positive numbers**, denoted $\mathbb{R}^+$, that satisfies the following two conditions:

(a) If $a, b \in \mathbb{R}^+$ then

$$a + b \in \mathbb{R}^+ \quad \text{and} \quad a \cdot b \in \mathbb{R}^+.$$

In other words, the positive numbers are closed under addition and multiplication.

(b) Given any real number a , one and only one of the following is true:

- (i) a is positive.
- (ii) a is zero.
- (iii) $-a$ is positive.

The complement of the set $\mathbb{R}^+ \cup \{0\}$, i.e. the set of numbers a that aren't positive or zero, is called the set of **negative numbers** and is denoted $\mathbb{R}^-$.

Definition 1.5. We define an order relation on $\mathbb{R}$ as follows: for all real numbers a and b we say that a is **less than** b ($a < b$) if $b - a \in \mathbb{R}^+$. In other words, $a < b$ means that the number $b - a$ is positive. For all real numbers, a and b , we say that $a \leq b$ if $b - a \in \mathbb{R}^+ \cup \{0\}$. In other words, $a \leq b$ means that the number $b - a$ is positive or zero. The symbols $>$ and $\geq$ are defined similarly.

16. Given real numbers $a, b \geq 0$ show that $a^2 \leq b^2$ if and only if $a \leq b$. Show that a similar result holds if the inequality is strict.

Proof. First, if $x, y \in \mathbb{R}^+ \cup \{0\}$ then $xy \in \mathbb{R}^+ \cup \{0\}$: if $x = 0$ or $y = 0$ then $xy = 0$, and otherwise $xy \in \mathbb{R}^+$ by Axiom 1.4(a).

($\Leftarrow$) Suppose $a \leq b$. By Definition 1.5, $b - a \in \mathbb{R}^+ \cup \{0\}$. Since $a, b \geq 0$, we have $b + a \in \mathbb{R}^+ \cup \{0\}$. By the field axioms,

$$b^2 - a^2 = (b - a)(b + a) \in \mathbb{R}^+ \cup \{0\},$$

so $a^2 \leq b^2$.

($\Rightarrow$) Suppose $a^2 \leq b^2$, and assume for contradiction that $a \leq b$ is false. Then $b - a \notin \mathbb{R}^+ \cup \{0\}$, so by Axiom 1.4(b), $-(b - a) = a - b \in \mathbb{R}^+$; that is, $b < a$.

Since $a = (a - b) + b$ with $a - b \in \mathbb{R}^+$ and $b \in \mathbb{R}^+ \cup \{0\}$, we get $a \in \mathbb{R}^+$, and then $a + b \in \mathbb{R}^+$. So

$$a^2 - b^2 = (a - b)(a + b) \in \mathbb{R}^+$$

by Axiom 1.4(a). But $a^2 \leq b^2$ means $-(a^2 - b^2) = b^2 - a^2 \in \mathbb{R}^+ \cup \{0\}$. Then both $a^2 - b^2$ and $-(a^2 - b^2)$ are positive or zero, contradicting the “one and only one” condition of Axiom 1.4(b). Hence $a \leq b$.

Strict version: $a^2 < b^2$ if and only if $a < b$.

($\Leftarrow$) Suppose $a < b$. Then $b - a \in \mathbb{R}^+$, and $b = (b - a) + a$ with $a \in \mathbb{R}^+ \cup \{0\}$, so $b \in \mathbb{R}^+$ and $b + a \in \mathbb{R}^+$. Therefore $b^2 - a^2 = (b - a)(b + a) \in \mathbb{R}^+$ by Axiom 1.4(a), so $a^2 < b^2$.

($\Rightarrow$) Suppose $a^2 < b^2$, and assume for contradiction that $a < b$ is false. Then $b - a \notin \mathbb{R}^+$, so by Axiom 1.4(b) either $b - a = 0$ or $a - b \in \mathbb{R}^+$. In either case $a - b \in \mathbb{R}^+ \cup \{0\}$, so $b \leq a$. By the non-strict result (with a and b swapped), $b^2 \leq a^2$, so $a^2 - b^2 \in \mathbb{R}^+ \cup \{0\}$. But $a^2 < b^2$ means $-(a^2 - b^2) \in \mathbb{R}^+$, contradicting Axiom 1.4(b). Hence $a < b$. $\square$

Definition 1.6. Given two distinct real numbers a and b such that $a < b$, we denote the set $\{x \in \mathbb{R} : a < x < b\}$ with the notation (a, b) . We denote $\{x \in \mathbb{R} : a \leq x \leq b\}$ with the notation $[a, b]$. In words (a, b) is the set of all real numbers between a and b . The set (a, b) is called an **open interval**. The set $[a, b]$ is called a **closed interval**. The values a and b are called the **endpoints**.

We will generalize the notation for intervals in the usual ways. For instance, $[a, b)$ is understood to mean the set $\{x \in \mathbb{R} : a \leq x < b\}$. We may also replace a with $-\infty$ or b with ∞ in which case our interval is **unbounded**. If both a and b are finite, the interval is **bounded**.

Definition 1.7. The **absolute value** of a real number x is denoted by $|x|$ and defined as

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}.$$

19. Show that for all $a, b \in \mathbb{R}$,

$$(a) \quad |a \cdot b| = |a| \cdot |b|$$

$$(b) \quad |a + b| \leq |a| + |b|$$

Proof of (b). By Problem 17(c) applied to ab , and then Problem 19(a),

$$ab \leq |ab| = |a| |b|, \quad \text{so} \quad 2ab \leq 2|a| |b|.$$

Using Problem 18 twice and adding $a^2 + b^2 = |a|^2 + |b|^2$ to both sides,

$$|a + b|^2 = (a + b)^2 = a^2 + 2ab + b^2 \leq |a|^2 + 2|a| |b| + |b|^2 = (|a| + |b|)^2.$$

By Problem 17(a), $|a + b| \geq 0$ and $|a| + |b| \geq 0$. Therefore Problem 16 applies, giving

$$|a + b| \leq |a| + |b|. \quad \square$$

20. Show that for $a, b, c \in \mathbb{R}$,

$$(a) \quad |a - b| \geq ||a| - |b||$$

$$(b) \quad |a - b| \leq |a - c| + |b - c|$$

Proof of (a). By Problem 19(b),

$$|a| = |(a - b) + b| \leq |a - b| + |b|, \quad \text{so} \quad |a| - |b| \leq |a - b|.$$

Swapping the roles of a and b gives $|b| - |a| \leq |b - a|$. By Problem 17(d), $|b - a| = |-(a - b)| = |a - b|$, so $-(|a| - |b|) \leq |a - b|$.

Let $x = |a| - |b|$. We have shown $x \leq |a - b|$ and $-x \leq |a - b|$. If $x \geq 0$, then $|x| = x \leq |a - b|$. If $x < 0$, then $|x| = -x \leq |a - b|$. In both cases, by Definition 1.7, $||a| - |b|| \leq |a - b|$. $\square$

21. Suppose that $x, a \in \mathbb{R}$ and $\epsilon > 0$ then

(a) $|x| < \epsilon$ if and only if $-\epsilon < x < \epsilon$

(b) $|x - a| < \epsilon$ if and only if $a - \epsilon < x < a + \epsilon$.

Proof of (a). First, if $p \leq q$ and $q < r$ then $p < r$: we have $r - p = (r - q) + (q - p)$ with $r - q \in \mathbb{R}^+$ and $q - p \in \mathbb{R}^+ \cup \{0\}$, so $r - p \in \mathbb{R}^+$ (directly if $q - p = 0$, and by Axiom 1.4(a) otherwise).

($\Rightarrow$) Suppose $|x| < \epsilon$. By Problem 17(c), $x \leq |x|$, so $x < \epsilon$. By Problems 17(c) and 17(d), $-x \leq |-x| = |x| < \epsilon$, so $-x < \epsilon$, and by Problem 11, $-\epsilon < x$. Hence $-\epsilon < x < \epsilon$.

($\Leftarrow$) Suppose $-\epsilon < x < \epsilon$. If $x \geq 0$, then $|x| = x < \epsilon$. If $x < 0$, then $|x| = -x$, and $-\epsilon < x$ gives $-x < \epsilon$ by Problem 11. In both cases $|x| < \epsilon$. $\square$

Proof of (b). Applying part (a) to the real number $x - a$,

$$|x - a| < \epsilon \iff -\epsilon < x - a < \epsilon.$$

For any $p, q \in \mathbb{R}$, $(q + a) - (p + a) = q - p$ by the field axioms, so $p < q$ if and only if $p + a < q + a$. Adding a to each part of the inequality gives

$$-\epsilon < x - a < \epsilon \iff a - \epsilon < x < a + \epsilon. \quad \square$$